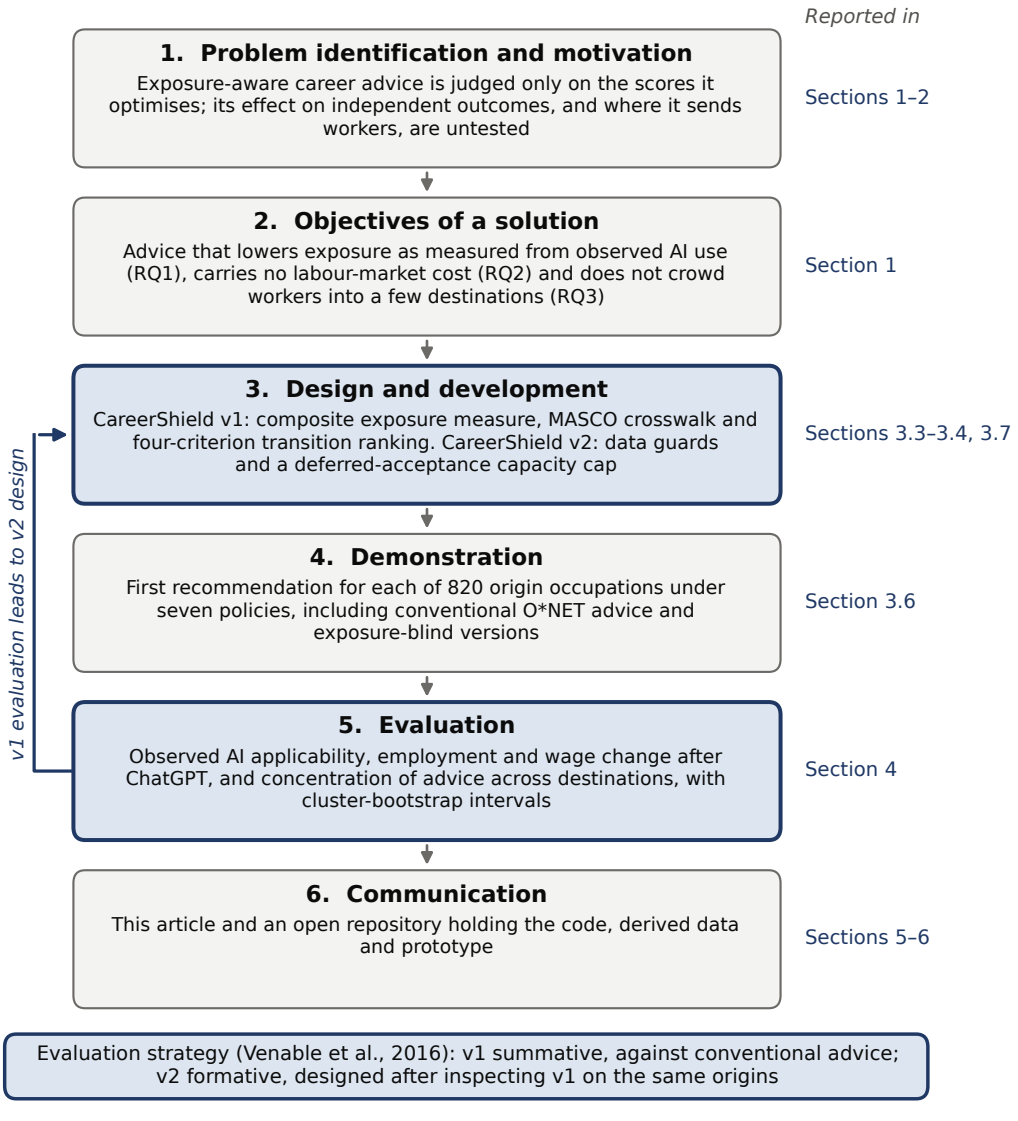
Reported in
1. Problem identification and motivation
Exposure-aware career advice is judged only on the scores it optimises; its effect on independent outcomes, and where it sends workers, are untested
Sections 1–2
2. Objectives of a solution
Advice that lowers exposure as measured from observed AI use (RQ1), carries no labour-market cost (RQ2) and does not crowd workers into a few destinations (RQ3)
Section 1
3. Design and development
CareerShield v1: composite exposure measure, MASCO crosswalk and four-criterion transition ranking. CareerShield v2: data guards and a deferred-acceptance capacity cap
Sections 3.3–3.4, 3.7
4. Demonstration
First recommendation for each of 820 origin occupations under seven policies, including conventional O*NET advice and exposure-blind versions
Section 3.6
5. Evaluation
Observed AI applicability, employment and wage change after ChatGPT, and concentration of advice across destinations, with cluster-bootstrap intervals
Section 4
6. Communication
This article and an open repository holding the code, derived data and prototype
Sections 5–6
v1 evaluation leads to v2 design
Evaluation strategy (Venable et al., 2016): v1 summative, against conventional advice; v2 formative, designed after inspecting v1 on the same origins